User Creation and Privilege Management

Module Title and Code: Introduction to Databases, UFCFTK-30-1

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TASK 3 (20%)

Creating Users and Granting Privileges (Based on Task 2 Tables)

Write SQL commands to:

1. Create a user data_analyst with the password Analyst@123.

Answer:

CREATE USER 'data_analyst'@'%' IDENTIFIED BY 'Analyst@123';

```
Setting environment for using XAMPP for Windows.
rauna@BATMAN c:\xampp
# mysql -u root -p
Enter password: ***********
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 11
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE tracker;
Database changed
MariaDB [tracker]> CREATE USER 'data_analyst'@'%' IDENTIFIED BY 'Analyst@123';
Query OK, 0 rows affected (0.009 sec)
```

In this task, a new user named **data_analyst** was implemented to the system with a **secure password**.
This user can access the database depending on the permission approved later in following tasks.

2. Grant the user data_analyst SELECT privileges on all tables in the TRACKER database.

Answer:

GRANT SELECT ON TRACKER.* TO 'data_analyst'@'%';

```
MariaDB [tracker]> GRANT SELECT ON TRACKER.* TO 'data_analyst'@'%';
Query OK, 0 rows affected (0.005 sec)
```

In this task, the user was granted permission to view the data from all the tables i.e. **AIRLINES** and **FLIGHT** inside the TRACKER database. However, no any modifications power was approved at this stage of the task.

3. Grant the user data_analyst INSERT privileges only on the FLIGHT table in the TRACKER database.

Answer:

GRANT INSERT ON TRACKER.FLIGHT TO 'data_analyst'@'%';

```
MariaDB [tracker]> GRANT INSERT ON TRACKER.FLIGHT TO 'data_analyst'@'%';
Query OK, 0 rows affected (0.004 sec)
```

In this task, the user was granted the permission to **add new records**, but only in the table named **FLIGHT**.
Other table i.e. **AIRLINES** remains restricted for the data modifications or insertion.

4. Revoke the INSERT privilege from the FLIGHT table.

Answer:

REVOKE INSERT ON TRACKER.FLIGHT FROM 'data_analyst'@'%';

```
MariaDB [tracker]> REVOKE INSERT ON TRACKER.FLIGHT FROM 'data_analyst'@'%';
Query OK, 0 rows affected (0.004 sec)
```

In this task, the previously assigned **INSERT** permission was **removed**.
This shows that how access rights can be adjusted according to the required condition.

Proof of Execution:

To confirm that everything worked correctly, the following tasks were performed:

1. Checking if user exists:

SELECT User, Host FROM mysql.user WHERE User = 'data_analyst';

```
MariaDB [tracker]> SELECT User, Host FROM mysql.user WHERE User = 'data_analyst';
+--------------+------+
| User         | Host |
+--------------+------+
| data_analyst | %    |
+--------------+------+
1 row in set (0.008 sec)
```

2. Checking granted privileges:

SHOW GRANTS FOR 'data_analyst'@'%';

```
MariaDB [tracker]> SHOW GRANTS FOR 'data_analyst'@'%';
+-----------------------------------------------------------------------------------------------------------------+
| Grants for data_analyst@%                                                                                       |
+-----------------------------------------------------------------------------------------------------------------+
| GRANT USAGE ON *.* TO `data_analyst`@`%` IDENTIFIED BY PASSWORD '*F7B87AEFE42254FF826AC55E6E9B92A04F7CD3CB' |
| GRANT SELECT ON `tracker`.* TO `data_analyst`@`%`                                                               |
+-----------------------------------------------------------------------------------------------------------------+
2 rows in set (0.001 sec)
```

These queries ensured that the user exists and that the correct permission were applied.

Tools and Technologies Used:

1. XAMPP Server

I used **XAMPP Server** to run a local **MySQL** environment on my system. I used its terminal to connect to localhost and perform all the required SQL Tasks, including **creating users**, **assigning privileges**, and **executing verification queries**.